STATISTICS · UNIT 4: STATISTICAL INFERENCE

The Later Start Investigation

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students build an evidence-based recommendation on a later school start time. They estimate support with a confidence interval, test whether most students support the change, and test whether students sleep less than the recommended 8 hours.

CP1 and Honors add t-procedures, sample size, two-proportion comparisons, a matched pairs pilot study, and power.

Standards Addressed

- S-IC.A.1 Inference about population parameters from random samples
- S-IC.B.4 Estimate a population mean or proportion; margin of error
- S-IC.B.5–6 Compare treatments; evaluate reports based on data
- AP Statistics: one- and two-sample z and t procedures, Type I/II errors, power

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: class vote and sleep survey. Part 1 confidence intervals.
Day 2	Part 2 significance tests.
Day 3	Part 3 errors and the board report; optional mock board meeting.

Materials

- Student handout and Evidence Cards
- Graphing calculators or a statistics app
- Optional: news articles on later start times

How the Three Versions Differ

Level	What changes
CP2	One-proportion interval with guided steps, hypotheses and a given p-value, and Type I and II errors in context.
CP1	Conditions, z-interval and test, sample size, t-interval and t-test, and a two-proportion interval.

Level	What changes
Honors	99% vs 95% intervals, two-proportion z-test, sample size for a difference, paired t-test and interval, and power.

Watch For (Common Misconceptions)

- Interpreting a 95% interval as “95% of students are in this range.”
- Saying the p-value is the probability H_0 is true.
- Using a two-sample test on paired data.
- Treating “fail to reject” as proof that H_0 is true.

Extension Ideas

- Collect a class sleep survey and run the t-test on your own data.
- Use an applet to see how power changes with n and α.
- Read a real study on school start times and identify its design.

Answer Key – CP2

Part 1: How Many Support It?

#	Task	Answer
1	Find the sample proportion $\hat{p}$.	$\hat{p} = 62 \div 100 = 0.62$
2	Find the standard error $\sqrt{(\hat{p}(1 - \hat{p}) \div n)}$.	$\sqrt{(0.62 \times 0.38 \div 100)} \approx 0.0485$
3	Find the 95% confidence interval $\hat{p} \pm 1.96 \cdot SE$.	$0.62 \pm 0.095 \rightarrow (0.525, 0.715)$
4	Interpret the interval in context.	We are 95% confident that between about 52.5% and 71.5% of all students at the school support the later start.

Part 2: Is It a Majority?

#	Task	Answer
5	Write the null and alternative hypotheses.	H_0: $p = 0.5$; H_a: $p > 0.5$ (p = proportion of all students who support)
6	The test statistic is $z = 2.4$ and the p-value is about 0.008. What does the p-value mean?	If only half of students supported it, there would be about a 0.8% chance of getting 62% or more support in a random sample of 100.
7	Use $\alpha = 0.05$ to write a conclusion.	$0.008 < 0.05$, so reject H_0; there is convincing evidence that a majority supports the later start.
8	Does the confidence interval agree with the test? Explain.	Yes; the whole interval (0.525, 0.715) is above 0.5.

Part 3: Could We Be Wrong?

#	Task	Answer
9	Describe a Type I error in this context.	Concluding a majority supports the change when actually it does not.
10	Describe a Type II error in this context.	Failing to find that a majority supports the change when actually it does.
11	Which error would be worse for the school board? Explain.	Answers vary; e.g., Type I, because the board would change the schedule for an unpopular plan.

Reflection

#	Prompt	Look for
R1	What does “95% confident” mean?	If we took many samples and made an interval from each, about 95% of the intervals would capture the true proportion.
R2	How would a bigger sample change the interval?	It would be narrower (smaller margin of error).

Answer Key – CP1

Part 1: Estimate Support

#	Task	Answer
1	Check the conditions (Random, 10%, Large Counts) for a one-proportion z-interval.	SRS ✓; 400 < 10% of students if the school has more than 4,000 (otherwise note the concern); 248 and 152 ≥ 10 ✓
2	Construct and interpret a 95% confidence interval.	$0.62 \pm 1.96(0.0243) = 0.62 \pm 0.048 \rightarrow (0.572, 0.668)$. We are 95% confident 57.2% to 66.8% of students support it.
3	How large a sample is needed for a margin of error of at most 3% at 95% confidence (use $\hat{p} = 0.5$)?	$n \geq (1.96 \div 0.03)^2 \times 0.25 \approx 1{,}067.1 \rightarrow 1{,}068$ students
4	Test H_0: $p = 0.5$ vs H_a: $p > 0.5$ at $\alpha = 0.05$. Give z, the p-value, and a conclusion.	$z = (0.62 - 0.5) \div \sqrt{(0.25 \div 400)} = 4.8$; p-value ≈ 0; reject H_0: convincing evidence of majority support.

Part 2: Are Students Short on Sleep?

#	Task	Answer
5	Construct a 95% t-interval for the mean sleep of all students (df = 15, $t^* \approx 2.131$).	$6.8 \pm 2.131(1.2 \div 4) = 6.8 \pm 0.64 \rightarrow (6.16, 7.44)$ hours
6	Test H_0: $\mu = 8$ vs H_a: $\mu < 8$. Find t and the p-value.	$t = (6.8 - 8) \div 0.3 = -4.0$; df = 15; p-value ≈ 0.0006
7	Write a conclusion in context. What condition about the data must you check?	Reject H_0: convincing evidence the mean is less than 8 hours. With n = 16, the sample data should show no strong skew or outliers.

Part 3: Errors and Grades

#	Task	Answer
8	Describe a Type I and a Type II error for the sleep test.	Type I: conclude students average less than 8 h when they do not. Type II: fail to find it when they really do sleep less.
9	Find $\hat{p}$ for each grade and the difference $\hat{p}_9 - \hat{p}_{12}$.	0.65 and 0.59; difference 0.06
10	Construct a 95% confidence interval for $p_9 - p_{12}$ and interpret it.	$0.06 \pm 1.96(0.0484) \rightarrow (-0.035, 0.155)$; it includes 0, so we cannot conclude the grades differ in support.
11	How would increasing α from 0.05 to 0.10 affect the chances of each error?	P(Type I) increases to 0.10; P(Type II) decreases (power increases).

Reflection

#	Prompt	Look for
R1	Why do we use t instead of z for the sleep data?	We do not know σ and estimate it with s, which adds variability.

#	Prompt	Look for
R2	Why does the grade comparison not prove the grades feel the same?	Failing to reject (or an interval with 0) means not enough evidence of a difference, not proof of no difference.

Answer Key – Honors

Part 1: Support for the Change

#	Task	Answer
1	Construct and interpret a 99% confidence interval for the proportion who support the later start.	$0.62 \pm 2.576(0.0243) = 0.62 \pm 0.063 \rightarrow (0.557, 0.683)$
2	Compare the 95% and 99% intervals. Why is one wider?	95%: (0.572, 0.668), 99% is wider because more confidence needs a larger critical value.
3	Carry out a two-proportion z-test of H_0: $p_9 = p_{12}$ vs H_a: $p_9 \neq p_{12}$ (pooled $\hat{p} = 0.62$).	$z = 0.06 \div \sqrt{(0.62 \times 0.38 \times (1/200 + 1/200))} \approx 1.24$; p-value ≈ 0.216; fail to reject H_0: no convincing evidence of a difference.
4	Find the sample size per grade needed to estimate $p_9 - p_{12}$ within ±0.05 at 95% (use 0.5 for each p).	$n \geq (1.96 \div 0.05)^2 \times (0.25 + 0.25) \approx 768.3 \rightarrow 769$ per grade

Part 2: Sleep and the Pilot Week

#	Task	Answer
5	Test H_0: $\mu = 8$ vs H_a: $\mu < 8$ for the sleep sample and state the conclusion.	$t = -4.0$, df = 15, p ≈ 0.0006; convincing evidence students average less than 8 hours.
6	Pilot week: find the mean and SD of the 10 differences (later start − normal).	$\bar{x}_d = 0.55$ h; $s_d \approx 0.468$ h
7	Carry out a paired t-test of H_0: $\mu_d = 0$ vs H_a: $\mu_d > 0$.	$t = 0.55 \div (0.468 \div \sqrt{10}) \approx 3.71$; df = 9; p ≈ 0.0024; convincing evidence students sleep more with the later start.
8	Construct a 95% interval for μ_d ($t^* \approx 2.262$) and interpret it in minutes.	$0.55 \pm 0.335 \rightarrow (0.215, 0.885)$ h, about 13 to 53 extra minutes of sleep

Part 3: Errors, Power, and Advice

#	Task	Answer
9	For the paired test, describe a Type II error and one consequence.	Concluding no extra sleep when the later start really helps; the board might reject a beneficial change.
10	Give two ways to increase the power of the pilot study.	Use more students; use a larger α; reduce variability (e.g., more careful sleep tracking); a longer pilot.
11	Explain why the pilot volunteers limit the conclusion and why the paired design was a good choice.	If volunteers were not random, results may not generalize; pairing compares each student to themselves, removing student-to-student variation.
12	Write a three-sentence recommendation to the board that states your evidence and its uncertainty.	Answers vary; should cite the support interval, the sleep test, the pilot interval, and limitations.

Reflection

#	Prompt	Look for
R1	Why is a paired t-test, not a two-sample t-test, correct for the pilot week?	The same students were measured twice, so the data are dependent pairs; we analyze the differences.
R2	How do a confidence interval and a significance test give different kinds of information to the board?	The test says whether there is an effect; the interval says how big it plausibly is.